UPSC GURUKUL

CLASS 8 - GEOGRAPHY

CHAPTER 5: INDUSTRIES

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- Define industry and explain its importance as a secondary activity.
- Classify industries on the basis of size, ownership and raw materials.
- Identify the main factors that influence the location of industries.
- Describe an industrial system in terms of inputs, processes and outputs.
- Locate and describe major industrial regions of the world and India.
- Explain the distribution and characteristics of the iron and steel, cotton textile and information technology industries.
- Understand the concept of industrial disaster and the need for safety measures.

What is an Industry?

meaning

blocks

p

Industry refers to an economic activity that is concerned with the production of goods, extraction of minerals or the provision of services. When we talk of industries in geography we usually mean secondary activities – the processing of raw materials into finished or semi-finished products.

def

Industry

An economic activity concerned with the production of goods, extraction of minerals or provision of services.

p

Industries transform raw materials into more valuable products. For example, iron ore is converted into steel, cotton into cloth, and wood into furniture. This process of value addition is the essence of industrial activity.

Remember

callout

Primary activities extract resources from nature. Secondary activities process those resources into manufactured goods. Tertiary activities provide services.

Classification of Industries

classification

blocks

p

Industries can be classified in several ways. The most common bases are size, ownership and the nature of raw materials used.

h3

1. On the Basis of Size

def

Small-scale industry

Uses small amounts of capital and technology. Usually run by individuals or small groups. Examples: basket weaving, pottery, handicrafts, and many cottage industries.

def

Large-scale industry

Uses large amounts of capital, advanced technology and employs a large number of workers. Examples: iron and steel plants, automobile factories, large textile mills.

h3

2. On the Basis of Ownership

list

items

• Private sector: Owned and operated by individuals or groups of individuals. Example: Reliance Industries, Tatas (in many sectors).

• Public sector: Owned and operated by the government. Example: Hindustan Aeronautics Limited, Steel Authority of India Limited (SAIL).

• Joint sector: Owned and operated by the state and individuals or a group of individuals. Example: Maruti Udyog Limited (originally).

• Cooperative sector: Owned and operated by the producers or suppliers of raw materials, workers or both. Example: Amul (dairy), sugar mills in Maharashtra and Uttar Pradesh owned by cooperative societies.

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3. On the Basis of Raw Materials

def

Agro-based industries

Use plant and animal products as raw materials. Examples: food processing, cotton textile, dairy products, leather industry.

def

Mineral-based industries

Use mineral ores as raw materials. Examples: iron and steel, cement, aluminium smelting.

def

Marine-based industries

Use products from the sea and oceans as raw materials. Examples: seafood processing, fish oil, salt from sea water.

def

Forest-based industries

Use forest produce as raw materials. Examples: paper, furniture, pharmaceuticals from medicinal plants, lac and resin industries.

An agro-based example

example

A cotton textile mill takes raw cotton (an agricultural product), spins it into yarn and weaves it into cloth. The value of the finished cloth is much higher than that of the raw cotton.

Factors Affecting Location of Industries

location-factors

blocks

p

Industries are not located everywhere. Their location is influenced by a combination of physical and human factors. The ideal location is one that minimises the cost of production and maximises profit.

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Important Locational Factors

list

items

• Availability of raw materials: Industries that use heavy or perishable raw materials are preferably located near the source of those materials (e.g., iron and steel near iron ore and coal mines).

• Power / energy: Continuous and cheap supply of power is essential. Many industries are located near coalfields or hydro-electric stations.

• Labour: Availability of skilled and unskilled workers at reasonable wages.

• Transport and communication: Good roads, railways, ports and communication facilities help in the movement of raw materials and finished goods.

• Market: Industries prefer locations close to large markets so that the finished products can be sold easily and transport costs are reduced.

• Water: Many industries (especially chemical, paper and iron & steel) need large quantities of water.

• Government policies: Governments may offer incentives, tax holidays or develop industrial estates to attract industries to particular regions.

• Capital and banking facilities: Availability of finance and banking services.

Key Idea

callout

No single factor decides the location of an industry. Usually a combination of several factors works together. Sometimes historical factors or the personal preference of an entrepreneur also play a role.

Industrial System

industrial-system

blocks

p

An industrial system consists of inputs, processes and outputs, just like a farm system.

list

items

• Inputs: Raw materials, labour, power, capital, machinery, land and water.

• Processes: The activities that convert the raw materials into finished products (e.g., spinning, weaving, smelting, assembling).

• Outputs: The finished product and any income earned. Sometimes there are by-products and waste materials as well.

Iron and steel industry as a system

example

Inputs: iron ore, coal, limestone, labour, capital, water and power. Processes: smelting in a blast furnace, refining, rolling. Outputs: steel (in various forms), slag (by-product) and income.

Remember

callout

The quality and cost of inputs and the efficiency of processes together determine the competitiveness of an industry.

Industrial Regions

industrial-regions

blocks

p

Industrial regions emerge when a number of industries locate close to each other and share the benefits of proximity – such as skilled labour, transport, banking and other facilities.

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Major Industrial Regions of the World

list

items

- Eastern North America
- Western and Central Europe
- Eastern Europe
- Eastern Asia (especially Japan, eastern China, South Korea)

h3

Major Industrial Regions of India

list

items

- Mumbai-Pune cluster
- Bangalore-Tamil Nadu region
- Hugli region (West Bengal)
- Ahmedabad-Vadodara region
- Chotanagpur industrial belt (Jharkhand–Odisha–West Bengal)
- Vishakhapatnam-Guntur belt
- Gurgaon-Delhi-Meerut region
- Kollam-Thiruvananthapuram belt in Kerala

p

The Chotanagpur region is one of the most important mineral-based industrial regions of India because of the rich deposits of coal, iron ore, manganese, bauxite and limestone.

Distribution of Major Industries

major-industries

blocks

h3

1. Iron and Steel Industry

p

The iron and steel industry is often called a basic or key industry because its products are used as raw materials by many other industries (automobiles, ship-building, construction, defence, etc.).

list

items

- Inputs: iron ore, coal (coking coal), limestone, manganese, labour, capital and water.
- Process: Iron ore is smelted in a blast furnace using coke and limestone. The molten iron is further refined to make steel.
- Major centres in the world: USA (Pittsburgh region), Germany (Ruhr), Russia, Ukraine, China, Japan.
- Major centres in India: Jamshedpur (Tata Steel), Bhilai, Durgapur, Rourkela, Bokaro, Burnpur, Bhadravati, Vijayanagar and Salem.

Why Jamshedpur?

example

Jamshedpur was chosen because of the availability of iron ore from Odisha and Jharkhand, coal from Jharia, water from the Subarnarekha river, and good rail connections. It is a classic example of an industry located near raw materials.

h3

2. Cotton Textile Industry

p

One of the oldest industries in India. It is agro-based and has both large-scale mills and a large decentralised handloom and power-loom sector.

list

items

- Raw material: cotton (and sometimes synthetic fibres).
- Major centres in India: Mumbai, Ahmedabad, Coimbatore, Kanpur, Chennai, Kolkata, Ludhiana and many centres in Maharashtra, Gujarat and Tamil Nadu.
- Earlier the industry was concentrated in Maharashtra and Gujarat because of the availability of raw cotton, humid climate, port facilities and capital. Later it spread to other parts of the country.

h3

3. Information Technology (IT) Industry

p

The IT industry deals with the storage, processing and distribution of information. It has become one of the fastest-growing industries in the world and in India.

list

items

- Main inputs: skilled human resource (software professionals), computers, internet connectivity and capital.
- Major centres in India: Bengaluru (often called the Silicon Valley of India), Hyderabad, Chennai, Pune, Noida, Gurgaon and Mumbai.
- The industry is less dependent on traditional locational factors such as raw materials and more dependent on the availability of skilled labour, good communication and a supportive environment.

Emerging Trend

callout

While traditional heavy industries still need to be near raw materials and power, modern industries such as

electronics and IT are more ‘footloose’ – they can locate where skilled labour and infrastructure are available.

Industrial Disaster and Safety

industrial-disaster

blocks

p

Industries can sometimes cause serious accidents that lead to loss of life, injury and environmental damage. These are called industrial disasters.

Bhopal Gas Tragedy (1984)

example

One of the world’s worst industrial disasters occurred in Bhopal when a poisonous gas (methyl isocyanate) leaked from a pesticide plant. Thousands of people died and many more suffered long-term health problems.

list

items

• Other examples include nuclear accidents (Chernobyl, Fukushima) and chemical plant explosions in different parts of the world.

• Causes often include poor safety standards, human error, lack of maintenance and inadequate emergency planning.

h3

Measures to Reduce Risk

list

items

• Strict safety regulations and regular inspection of plants.

• Proper training of workers.

• Storage of hazardous materials away from residential areas.

• Clear emergency plans and warning systems.

• Use of cleaner technologies and better waste management.

Chapter Takeaway

callout

Industries are essential for economic development and employment, but their location must be carefully planned and their operations must follow strict safety and environmental standards so that development remains sustainable.

takeaways

• Industry is a secondary activity that processes raw materials into finished goods.

• Industries are classified by size (small-scale / large-scale), ownership (private, public, joint, cooperative) and raw materials (agro-based, mineral-based, marine-based, forest-based).

• Location of industries depends on raw materials, power, labour, transport, market, water and government policies.

• An industrial system has inputs, processes and outputs.

• Major industrial regions develop where several industries cluster together.

• Iron and steel is a basic industry; cotton textile is a leading agro-based industry; IT is a fast-growing modern industry.

• Industrial accidents can be catastrophic; strict safety measures are essential.

Industries process raw materials into more valuable products and form the secondary sector of the economy. They can be classified according to size, ownership and the type of raw material used. The location of an

industry is decided by a combination of factors such as raw materials, power, labour, transport, market and government policy. An industrial system consists of inputs, processes and outputs. Important industries studied include iron and steel, cotton textiles and information technology. While industries bring development and employment, they must be operated with high safety standards to prevent disasters and environmental damage.

flashcards

What is an industry?

An economic activity concerned with the production of goods, extraction of minerals or provision of services.

Name the four types of industries on the basis of ownership.

Private sector, public sector, joint sector and cooperative sector.

What are agro-based industries?

Industries that use plant and animal products as raw materials.

Why is the iron and steel industry called a basic industry?

Because its products are used as raw materials by many other industries.

Name two important factors that affect the location of industries.

Availability of raw materials and power / energy (any other valid factors also acceptable).

Which city is often called the Silicon Valley of India?

Bengaluru (Bangalore).

Name one major iron and steel plant in India.

Jamshedpur (Tata Steel) or Bhilai, Bokaro, Rourkela, Durgapur, etc.

What is a joint-sector industry?

An industry owned and operated jointly by the government and private individuals or groups.

Give one example of a cooperative sector industry in India.

Amul (dairy) or many sugar mills owned by cooperative societies.

Why must industries follow strict safety measures?

To prevent industrial disasters that can cause loss of life, injury and environmental damage.

q1

Industry is mainly regarded as a:

options

- Primary activity
- Secondary activity
- Tertiary activity
- Quaternary activity

1

Manufacturing industries process raw materials into finished goods and therefore belong to the secondary sector.

meaning

q2

Which of the following is a large-scale industry?

options

- Basket weaving
- Pottery

- Iron and steel plant
- Handloom weaving

2

Iron and steel plants require huge capital, advanced technology and employ large numbers of workers.

classification

q3

Amul is an example of a:

options

- Public sector industry
- Private sector industry
- Cooperative sector industry
- Joint sector industry

2

Amul is owned and operated by a cooperative of milk producers.

classification

q4

Industries that use mineral ores as raw materials are called:

options

- Agro-based
- Mineral-based
- Marine-based
- Forest-based

1

Mineral-based industries depend on mineral ores such as iron ore, bauxite, etc.

classification

q5

Which factor is most important for the location of an iron and steel plant?

options

- Nearness to a large market only
- Availability of iron ore and coking coal
- Availability of skilled IT professionals
- Cold climate

1

Iron and steel plants are usually located near the sources of heavy raw materials (iron ore and coal) to reduce transport costs.

location-factors

q6

Which of the following is an input in an industrial system?

options

- Finished steel
- Profit
- Iron ore and coal

- Sales revenue

2

Raw materials such as iron ore and coal are inputs; finished products and profit are outputs.

industrial-system

q7

The Chotanagpur industrial region is important mainly because of:

options

- Large IT companies
- Rich mineral deposits
- Tea plantations
- Cotton textile mills only

1

The region has abundant coal, iron ore and other minerals that support heavy industries.

industrial-regions

q8

Which city is known as the ‘Silicon Valley of India’?

options

- Mumbai
- Kolkata
- Bengaluru
- Chennai

2

Bengaluru is the leading centre of the information technology industry in India.

major-industries

q9

The iron and steel industry is called a basic industry because:

options

- It is the oldest industry
- Its products are used by many other industries
- It employs the largest number of workers
- It is found only in developed countries

1

Steel is a basic raw material for automobiles, construction, machinery, defence and many other industries.

major-industries

q10

Which of the following is an agro-based industry?

options

- Iron and steel
- Cotton textile
- Aluminium smelting
- Cement

1

Cotton textile industry uses cotton, an agricultural product, as its main raw material.

classification

q11

Tata Steel plant is located at:

options

- Bhilai
- Rourkela
- Jamshedpur
- Bokaro

2

The first large integrated steel plant of India was set up by the Tatas at Jamshedpur.

major-industries

q12

Which of the following is a public sector undertaking?

options

- Reliance Industries
- Tata Motors
- Steel Authority of India Limited (SAIL)
- Amul

2

SAIL is owned and operated by the Government of India.

classification

q13

An industrial region develops when:

options

- Only one large factory is set up
- Several industries locate close to one another and share facilities
- Farming becomes the main activity
- The area has no mineral resources

1

Clustering of industries creates shared advantages of labour, transport, banking and infrastructure.

industrial-regions

q14

Which of the following is a major cotton textile centre in India?

options

- Jamshedpur
- Bhilai
- Ahmedabad
- Rourkela

2

Ahmedabad has traditionally been one of the leading cotton textile centres of India.

major-industries

q15

The Bhopal gas tragedy is an example of:

options

- A natural disaster
- An industrial disaster
- A biological disaster
- A transport accident

1

It was caused by the leakage of poisonous gas from an industrial plant.

industrial-disaster

q16

Which of the following industries is least dependent on the location of raw materials?

options

- Iron and steel
- Cement
- Information technology
- Sugar

2

IT industry mainly needs skilled manpower and good communication facilities rather than bulky raw materials.

major-industries

q17

Marine-based industries use raw materials obtained from:

options

- Forests
- Mines
- Seas and oceans
- Farms

2

Marine-based industries process products such as fish, seafood, salt from sea water, etc.

classification

sourceNote

These are expanded study notes aligned to the Class 8 Geography chapter topic Industries. Use your prescribed NCERT textbook and teacher’s instructions as the final authority for exact textbook wording and syllabus coverage.